

ILLUSTRATED PARTS CATALOG (IPC)

BOEING 747-8F FLEET

DOCUMENT NUMBER: B748-IPC-002

VOLUME 1: AIRFRAME, SYSTEMS, AND POWERPLANT

WARNING: THIS DOCUMENT CONTAINS SYNTHETIC TEST DATA
AUTHORIZED FOR HACKATHON AND PROTOTYPING USE ONLY.
DO NOT USE FOR ACTUAL AIRCRAFT MAINTENANCE.

INTRODUCTION

This Illustrated Parts Catalog (IPC) is compiled in accordance with the Air Transport Association (ATA) Specification 100. The purpose of this manual is to assist maintenance, engineering, and provisioning personnel in identifying aircraft replacement parts and determining their installation effectivity across the 747-8F fleet.

How to Use This Catalog

When referencing a component in this catalog, locate the appropriate Air Transport Association (ATA) chapter. The illustrations (Figures) contain callout numbers (Items) which correspond directly to the Detailed Parts List on the same or subsequent page.

For example, if you are looking for an O-ring on the pack flow control valve, navigate to ATA Chapter 21, locate the Exploded View Figure, identify the callout number pointing to the O-ring, and match that number in the **ITEM** column of the adjacent parts list.

Understanding Effectivity Codes

The Effectivity (EFF) column in the Detailed Parts List indicates the specific aircraft configuration or line number for which the part is applicable. If the EFF column is left blank, the part applies universally to all aircraft models and block configurations covered by this specific catalog.

- **ALL:** Applies to the entire 747-8F fleet without restriction.
- **8F001-8F099:** Early production block airframes.
- **8F100-8F199:** Late production block featuring the strengthened wingbox structure.

Alternate Parts Classification

This catalog strictly categorizes alternate parts to ensure compliance with FAA Airworthiness Directives and Boeing Engineering specifications. Unauthorized substitutions are strictly prohibited.

- **True Alternate:** 100 percent cross-compatible. Form, fit, and function identical to primary.
- **Oversized Version:** Designed for structural wear. Requires SRM consultation and specific hardware kits.
- **Optional Fit:** Conditionally approved (e.g., weight restrictions, emergency ferry flights). Requires Engineering Liaison (EL) sign-off.

VENDOR CAGE CODES

The Commercial and Government Entity (CAGE) code is a five-character ID number used extensively to identify vendors and suppliers in the aviation supply chain. This code is often stamped directly onto the part alongside the Part Number (PN) for traceability.

CAGE	VENDOR NAME & LOCATION
81205	BOEING COMPANY, SEATTLE WA
99207	GE AVIATION, EVENDALE OH
73030	PARKER HANNIFIN, IRVINE CA
59364	COLLINS AEROSPACE, CHARLOTTE NC
06303	HONEYWELL INT, PHOENIX AZ
14277	WOODWARD INC, ROCKFORD IL
97384	SAFRAN LANDING SYS, VELIZY FR
51234	MEGGITT SENSORS, COVENTRY UK
33091	EATON AEROSPACE, CLEVELAND OH
11111	MOOG INC, EAST AURORA NY
22222	BAE SYSTEMS, ENDICOTT NY

Standard Abbreviations

To conserve space in the nomenclature column of the Detailed Parts Lists, standard abbreviations are utilized throughout this catalog.

- **ASSY:** Assembly
- **BRKT:** Bracket
- **CPLG:** Coupling
- **FIG:** Figure
- **HDWR:** Hardware
- **INBD:** Inboard
- **MTOW:** Maximum Takeoff Weight
- **OUTBD:** Outboard
- **OVR:** Oversized
- **PN:** Part Number
- **QTY:** Quantity Per Assembly
- **VLV:** Valve

1 CHAPTER 21 - AIR CONDITIONING

1.1 21-50-01: AIR CYCLE MACHINE PACK VALVE

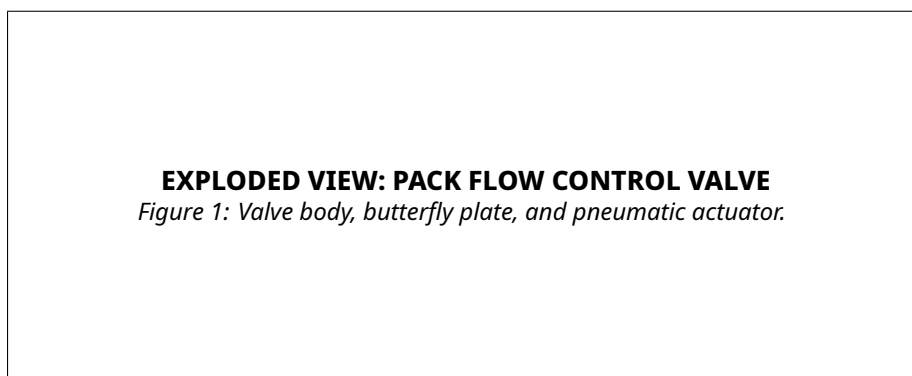


Figure 1: Air Cycle Machine (ACM) Pack Valve

Detailed Parts List (FIG 1)

ITEM	PART NUMBER	EFF	NOMENCLATURE	CLASSIFICATION / NOTES
01	VLV-ACM-990	ALL	VALVE ASSY - PACK	Primary Part. Status: Active.
01A	VLV-ACM-991-ALT	ALL	VALVE ASSY - PACK	True Alternate. PC software update recommended.
01B	VLV-ACM-HA	ALL	VALVE ASSY - HIGH ALT	Optional Fit. Approved for above 8,000 ft MSL.
02	SL-990-1	ALL	SEAL - O-RING	Replace every time valve is removed.
03	WSH-44A	ALL	WASHER - FLAT	Use under bolt head.
04	NUT-88-LOCK	ALL	NUT - SELF LOCKING	Discard after single use.
05	SCR-8-32-05	ALL	SCREW - RETAINING	Torque to 15 in-lbs.

2 CHAPTER 24 - ELECTRICAL POWER

2.1 24-20-05: INTEGRATED DRIVE GENERATOR (IDG)



Figure 2: Integrated Drive Generator

Detailed Parts List (FIG 2)

ITEM	PART NUMBER	EFF	NOMENCLATURE	CLASSIFICATION / NOTES
01	IDG-747-90K	ALL	GENERATOR ASSY - IDG	Primary Part. 90kVA output.
01A	IDG-747-90K-R	ALL	GENERATOR ASSY - IDG	True Alternate. Remanufactured unit.
02	CLP-IDG-V	ALL	CLAMP - V-BAND	Quick disconnect clamp.
03	PIN-QDC-1	ALL	PIN - SHEAR	Replace if generator disconnects in flight.
04	O-R-IDG-02	ALL	O-RING - COOLING LINE	Qty 2 required per installation.

3 CHAPTER 27 - FLIGHT CONTROLS

3.1 27-11-05: INBOARD AILERON POWER CONTROL UNIT

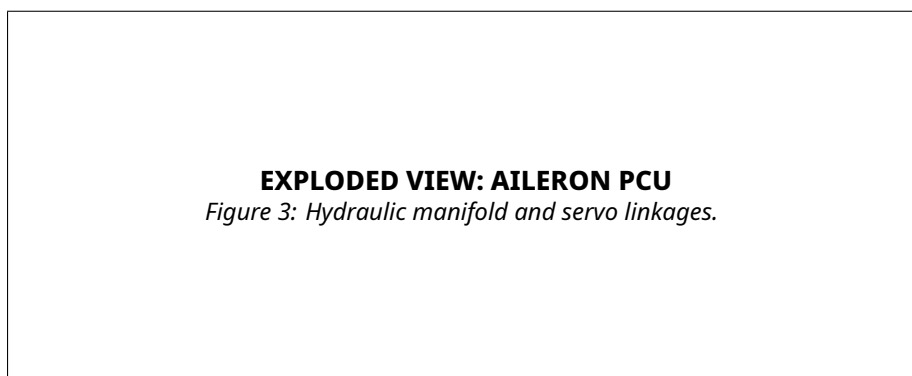


Figure 3: Inboard Aileron Power Control Unit

Detailed Parts List (FIG 3)

ITEM	PART NUMBER	EFF	NOMENCLATURE	CLASSIFICATION / NOTES
01	PCU-AIL-INB-01	ALL	PCU ASSY - INBD	Primary Part.
01A	PCU-AIL-INB-02A	ALL	PCU ASSY - INBD	True Alternate. Direct fit replacement.
01B	PCU-AIL-INB-03-OS	ALL	PCU ASSY - OVERSIZE	Oversized Version. Requires re-reaming of lug.
02	FL-112	ALL	FILTER - HYD	10-micron return filter.
03	BLT-AIL-7	ALL	BOLT - TITANIUM	Upper attach bolt. Do not substitute.
04	NUT-AIL-7	ALL	NUT - CASTELLATED	Install with cotter pin (Item 05).
05	PIN-COT-01	ALL	PIN - COTTER	Stainless steel. Single use.

4 CHAPTER 28 - FUEL SYSTEMS

4.1 28-22-10: CENTER WING TANK BOOST PUMP

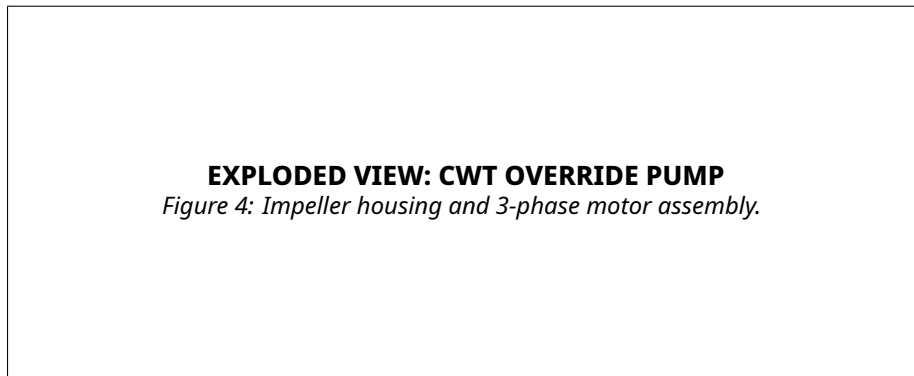


Figure 4: Center Wing Tank Boost Pump

Detailed Parts List (FIG 4)

ITEM	PART NUMBER	EFF	NOMENCLATURE	CLASSIFICATION / NOTES
01	PMP-CWT-88A	ALL	PUMP ASSY - OVERRIDE	Primary Part.
01A	PMP-CWT-88B	ALL	PUMP ASSY - OVERRIDE	True Alternate. Features upgraded shaft seal.
02	GSK-CWT-1	ALL	GASKET - PUMP MOUNT	Single-use crush gasket.
03	SCR-10-32	ALL	SCREW - MACHINE	Mounting screws (Qty 8).
04	LKW-10	ALL	WASHER - LOCK	Qty 8. Ensure correct seating.

5 CHAPTER 29 - HYDRAULIC POWER

5.1 29-11-00: ENGINE DRIVEN PUMP (EDP)

EXPLODED VIEW: ENGINE DRIVEN PUMP

Figure 5: Variable displacement pump housing.

Figure 5: Hydraulic Engine Driven Pump

Detailed Parts List (FIG 5)

ITEM	PART NUMBER	EFF	NOMENCLATURE	CLASSIFICATION / NOTES
01	EDP-3000-A	ALL	PUMP ASSY - HYDRAULIC	Primary Part. 3000 PSI output. True Alternate. Fully interchangeable.
01A	EDP-3000-B	ALL	PUMP ASSY - HYDRAULIC	
02	GSK-EDP-SPL	ALL	GASKET - SPLINE	Metallic crush gasket.
03	FIT-HYD-08	ALL	FITTING - PRESSURE	1/2 inch pressure outlet fitting.
04	FLT-HYD-CASE	ALL	FILTER - CASE DRAIN	Replace filter on every EDP change.

6 CHAPTER 32 - LANDING GEAR

6.1 32-30-01: MAIN LANDING GEAR RETRACTION ACTUATOR



Figure 6: Main Landing Gear Retraction Actuator Assembly

Detailed Parts List (FIG 6)

ITEM	PART NUMBER	EFF	NOMENCLATURE	CLASSIFICATION / NOTES
01	ACT-747-PRM	ALL	ACTUATOR ASSY - MLG	Primary Part. Status: Active.
01A	ACT-747-ALT1	ALL	ACTUATOR ASSY - MLG	True Alternate. Direct fit. 100 per- cent cross-compatible.
01B	ACT-747-OVR	ALL	ACTUATOR ASSY - OVR	Oversized Version. Requires Bracket B-99.
01C	ACT-747-OPT	ALL	ACTUATOR ASSY - OPT	Optional Fit. Ferry flights only. MTOW 850k lbs.
02	B-99	ALL	BRACKET - STRUCTURAL	Required only with ACT-747-OVR.
03	HK-22	ALL	HARDWARE KIT - OVR	Titanium bolts for use with B-99.
04	PN-88-TR	ALL	PIN - TRUNNION ATTACH	Standard lower attach pin.
05	C-CLIP-7	ALL	RING - RETAINING	Secures trunnion pin.
06	LUBE-FIT-01	ALL	FITTING - GREASE	Zerk style lubrication fitting.

7 CHAPTER 49 - AIRBORNE AUXILIARY POWER

7.1 49-10-22: APU STARTER GENERATOR

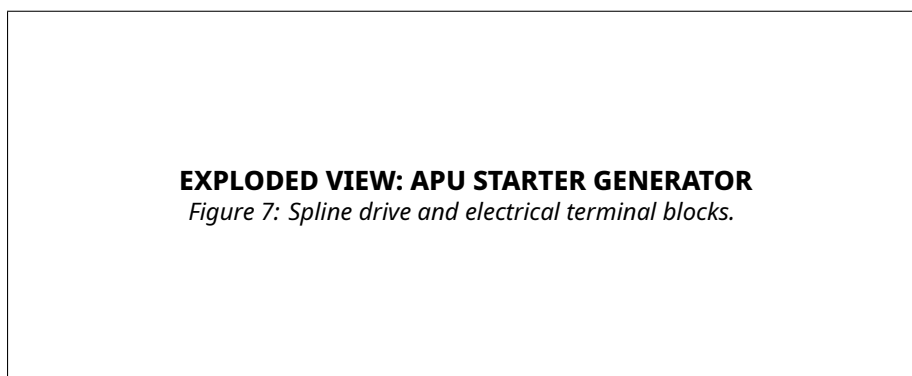


Figure 7: APU Starter Generator (90kVA)

Detailed Parts List (FIG 7)

ITEM	PART NUMBER	EFF	NOMENCLATURE	CLASSIFICATION / NOTES
01	SG-APU-90kVA	ALL	STARTER/GEN - APU	Primary Part. 90 kVA output.
01A	SG-APU-120kVA-ALT	ALL	STARTER/GEN - HD	Optional Fit. 120 kVA output. Requires ST-992 mod.
02	ST-992	ALL	STRUT - AFT COWL	Modified strut required for 120kVA clearance.
03	CBL-APU-G	ALL	CABLE - GROUND STRAP	Braided grounding strap.
04	LUG-APU-P	ALL	TERMINAL - LUG	Phase connection lug (Qty 3).

8 CHAPTER 73 - ENGINE FUEL AND CONTROL

8.1 73-11-05: MAIN ENGINE FUEL PUMP (GENx-2B)

EXPLODED VIEW: MAIN ENGINE FUEL PUMP

Figure 8: Dual-stage centrifugal pump assembly.

Figure 8: Main Engine Fuel Pump

Detailed Parts List (FIG 8)

ITEM	PART NUMBER	EFF	NOMENCLATURE	CLASSIFICATION / NOTES
01	PMP-GENX-MFP1	ALL	PUMP ASSY - MAIN	Primary Part. Active status.
01A	PMP-GENX-MFP2	ALL	PUMP ASSY - MAIN	True Alternate. Fully interchangeable.
02	SPL-99	ALL	SPLINE DRIVE CPLG	Ensure lubrication prior to installation.
03	O-R-GENX-4	ALL	PACKING - O-RING	Main housing seal.
04	STUD-GE-8	ALL	STUD - MOUNTING	Threaded mounting stud. Qty 6.
05	NUT-GE-8	ALL	NUT - SELF LOCKING	High temperature alloy nut. Qty 6.

APPENDIX A: NUMERICAL INDEX

Use this alphabetical and numerical index to cross-reference specific Part Numbers to their respective chapters, sections, and figures within this Illustrated Parts Catalog volume.

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ACT-747-ALT1	32-30-01	9
ACT-747-OPT	32-30-01	9
ACT-747-OVR	32-30-01	9
ACT-747-PRM	32-30-01	9
B-99	32-30-01	9
BLT-AIL-7	27-11-05	6
C-CLIP-7	32-30-01	9
CBL-APU-G	49-10-22	10
CLP-IDG-V	24-20-05	5
EDP-3000-A	29-11-00	8
EDP-3000-B	29-11-00	8
FIT-HYD-08	29-11-00	8
FL-112	27-11-05	6
FLT-HYD-CASE	29-11-00	8
GSK-CWT-1	28-22-10	7
GSK-EDP-SPL	29-11-00	8
HK-22	32-30-01	9
IDG-747-90K	24-20-05	5
IDG-747-90K-R	24-20-05	5
LKW-10	28-22-10	7
LUBE-FIT-01	32-30-01	9
LUG-APU-P	49-10-22	10
NUT-88-LOCK	21-50-01	4
NUT-AIL-7	27-11-05	6
NUT-GE-8	73-11-05	11
O-R-GENX-4	73-11-05	11
O-R-IDG-02	24-20-05	5
PCU-AIL-INB-01	27-11-05	6
PCU-AIL-INB-02A	27-11-05	6
PCU-AIL-INB-03-OS	27-11-05	6
PIN-COT-01	27-11-05	6
PIN-QDC-1	24-20-05	5
PMP-CWT-88A	28-22-10	7
PMP-CWT-88B	28-22-10	7
PMP-GENX-MFP1	73-11-05	11
PMP-GENX-MFP2	73-11-05	11
PN-88-TR	32-30-01	9
SCR-10-32	28-22-10	7
SCR-8-32-05	21-50-01	4
SG-APU-120kVA-ALT	49-10-22	10
SG-APU-90kVA	49-10-22	10
SL-990-1	21-50-01	4
SPL-99	73-11-05	11
ST-992	49-10-22	10
STUD-GE-8	73-11-05	11
VLV-ACM-990	21-50-01	4
VLV-ACM-991-ALT	21-50-01	4
VLV-ACM-HA	21-50-01	4
WSH-44A	21-50-01	4

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